

Saarathi — Solution Summary

Expedia Group Campus Hackathon 2026 — Innovation Round Problem Statement 1: AI Air Travel Companion

I. The Selected Problem Statement

AI Air Travel Companion — building an AI-powered air travel assistant that suggests and optimizes flight options based on user profiles and inferred travel preferences, supporting multi-city travel, direct-versus-connecting trade-offs, flexible date constraints, and seasonal pricing patterns.

Within this problem statement, Saarathi focuses on two of the toolkit's "strong solution" dimensions most deliberately: **trade-off awareness and explanation**, and **intelligent preference extraction from raw, unstructured user data** — treating these not as features alongside a recommender, but as the recommender's entire reason for existing.

II. The User / Business Problem Being Addressed

Booking a flight is not a search problem — it is a decision problem. A traveler is silently weighing cost, time, fatigue, flexibility, airline loyalty, baggage, and arrival windows, and no existing tool helps them *decide*. Sites like Kayak or Expedia itself respond to this by surfacing more rows and more filters — more information, never more judgment. The traveler still does all of the deciding alone, and once they book, they're never told what they gave up to get there.

The gap isn't a lack of data. It's the absence of a system that takes a position, defends it with evidence, and is honest about exactly when it would be wrong.

III. The Proposed Solution

Saarathi is an AI travel strategist built around one screen — the **Decision Screen** — that a judge or traveler understands in ten seconds, without narration. It answers four questions for every recommendation, not one:

1. **What should I book?** — a single, confident verdict (not a list of ten).
2. **Why is this the best decision?** — an evidence trail quoting the traveler's own words, tagged strong or weak based on whether structured profile data and behavioral history agree.
3. **What am I giving up?** — an opportunity-cost panel showing what the cheapest, fastest, most flexible, most comfortable, and most date-flexible alternatives would have cost or saved instead.
4. **What would change the AI's mind?** — the flagship feature. Because ranking is computed with a transparent, linear weighted-scoring formula (not an opaque model), Saarathi can solve that formula algebraically to state the exact price at which a

runner-up flight would win — "United 88 becomes the pick if its fare drops below \$543" — and can flip the winner live in front of the user by tapping that exact number.

Underneath, the architecture keeps a strict boundary: a **deterministic core** (rule-based and embedding-based preference extraction, weighted scoring, closed-form counterfactual algebra, branch-and-bound multi-city routing) does every decision, while an **LLM (Groq/Llama 3.3, via LangChain)** touches the system only at two narrow, replaceable edges — phrasing the final explanation, and parsing free-text requests into structured origin/destination — never deciding a ranking or a number. Every AI touchpoint has a deterministic fallback, so the app never breaks live.

IV. Expected Value or Impact

For travelers, Saarathi replaces "here are 40 flights, good luck" with a confident recommendation, an honest account of the trade-off being made, and a clear, numeric answer to "what would it take for me to choose differently" — the kind of clarity a good human travel agent provides, at the speed and scale a search engine requires.

For Expedia Group, this reframes flight recommendation from a ranking problem into a **trust and decision-support problem** — the same shift that made Google Maps ("fastest, cheapest, avoids tolls") more useful than a plain list of routes. A recommendation system that can explain its own decision boundary is more defensible to a skeptical customer, more auditable for internal review, and more differentiated against competitors who can only say "this is what we recommend," not "and here is exactly what would change that."